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This study was an unfunded project. Dr. Gailey and Dr. Kim are involved and/or stand to gain commercially from the development of this product. The results of this study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by ACSM.

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Abstract

Purpose: To determine if a measure of lower limb segment stability derived from body-worn inertial measurement units (IMU) can predict risk for lower limb musculoskeletal injury in Division I Collegiate Football Players (D1CFP).

Methods: The Region of Limb Stability (ROLS) values were collected in a cohort of D1CFPs during pre-season. ROLS is a measure of knee joint stability, defined by thigh and shank excursion (cm^2) in the anterior-posterior (AP) and medial-lateral (ML) direction during single limb stance (SLS). The ROLS Symmetry Index (SI) (%) is the ratio between lower limb ROLS values where 100% suggests absolute symmetry.

Results: One-hundred four D1CFP participated in this study and were divided into two groups: 1) No previous lower limb injury or no in-season injury (n = 70, "non-injured group") and 2) No previous lower limb injury, but in-season injury requiring surgery (n = 34, "injured group" group). The mean ± standard deviation (SD) ROLS SIs was 82.86 ± 14.75% and 65.58 ± 16.46% for the non-injured and injured group, respectively. Significant differences in ROLS SI were found between groups ($p < 0.001$). The ROLS SI demonstrated an AUC of 0.8 ($p < 0.001$; 95% CI: 0.71 -0.88) with a SE of 0.04 indicating that the ROLS SI has good predictive accuracy in detecting those healthy D1CFPs at risk for lower limb injury resulting in surgery.

Conclusion: The ROLS SI was found to have good predictive accuracy in detecting individuals at risk for injury that were healthy and asymptomatic during preseason testing. Increase in thigh and shank excursions and/or decrease in symmetry index (SI) between lower limbs may be a predictor of risk for future injury.

Keywords: Region of Limb Stability, Injuries, Relative Risk, Lower Limb, Surgery, Single Limb Stance

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Introduction

Collegiate and professional-level sport athletes are vulnerable to lower limb musculoskeletal (MSK) injuries, particularly at the knee and ankle (1-2). Lower limb MSK injuries have been found to significantly impact the ability to return to sport (2). Walden et al. found that only 65% of European men's professional football players with anterior cruciate ligament (ACL) injuries were able to return to their pre-injury performance levels after reconstruction (3). Moreover, it may take up to two years following ACL-reconstruction (ACL-R) for a lower limb to return to pre-injury strength (4). Not only is achieving prior levels of function after an ACL injury a challenge but athletes are also at a higher risk for re-injury, injury to the contralateral limb, and have an increased risk for associated chronic conditions (1,2,5-8).

Sensitive and specific objective tests and measures for detecting which athletes are at-risk for musculoskeletal injuries remain elusive in sports medicine (9-15). As a result, diagnostic imaging such as Magnetic Resonance Imaging (MRI), three-dimensional motion analysis (3DMA) systems, and force plates have been utilized to assess risk for lower limb injury (9-11). Although these measures have clinical utility, most are not practical for clinicians to use due to high costs and the need for special equipment.

Since no consensus exists for determining a standard for screening measures of lower limb injury risk, researchers and clinicians have combined multiple clinical factors and measures to address this challenge (12-15). Performance based outcome measures (PBOM) such as the single-legged hop test, coupled with standard clinical measures such as muscle strength evaluations and range of motion measures, have been employed to assess limb asymmetry and risk for re-injury

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(12). To determine probability of injury during soccer season, Liporaci et al., used a regression model with a concordance index of 84% to propose a nomogram that assigns a point value to individual physical and sport characteristics (14). Similarly, Myer et al., proposed a nomogram with 84% sensitivity and 67% specificity that examines tibial length, knee valgus motion, knee flexion range of motion, body mass, and hamstring to quadriceps ratio to predict knee abduction moments (15). Although nomograms may be promising, limitations such as lack of utility and expediency in a sports medicine or rehabilitation setting have been noted (12-15).

To address limitations of current approaches, Region of Limb Stability (ROLS) was developed. The ROLS is a measure of knee joint stability, defined by thigh and shank excursion (cm^2) in the anterior-posterior (AP) and medical-lateral (ML) direction during single limb stance (SLS) (16). The ROLS is calculated using a mobile lower limb motion capture system called CaneSense™, which consists of four wireless IMUs secured in two elastic knee sleeves (two IMUs per knee sleeve) that communicate independently via Bluetooth Low Energy (BLE) to a custom software application on a mobile tablet that processes the kinematic data in real-time and transmits the data to a secure cloud database at the University of Miami (16). The custom designed wireless IMUs and software are compact (2.5 cm×2.5 cm×1.25 cm each) and contains an accelerometer, gyroscope, and magnetometer, as well as a Bluetooth Low Energy transceiver. All data, i.e. 3-axis linear accelerations (±160 m/s2 at 10-bit resolution), 3-axis angular velocities (±2000 deg/s at 16-bits), and 3-axis magnetic fields (±2500 nT at 13-bits) are sampled at 50 Hz, transmitted to a custom iOS app, and then stored on a central PostgreSQL database at the University of Miami Center for Computational Sciences. In-flight transmission and database storage are secured and encrypted (16).

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The position of the IMUs on the elastic knee sleeves are at the medial tibial flare and the lateral proximal thigh, just anterior to the insertion of the iliotibial band. The positions were chosen in order to avoid contracting musculature on the stance limb during SLS (16,17). After donning the knee sleeves and sensor system, the participant performs an alignment trial which consists of a sit-stand activity followed by walking 10 steps. The alignment trial virtually aligns the sensors with the knee joint axis. During SLS, the ROLS excursion values can be expressed in numeric, 2 and 3-dimensional graphical formats and is comparable to other forms of 3DMA (16). Then, segmental excursions are computed using aligned acceleration data from the two IMUs donned on the supporting limb during SLS (Figure 1(a)). Specifically, two iterations of integration and high pass filtering at 0.3Hz were applied to eliminate drift and calculate an estimate of IMU displacement on the supporting limb during SLS (16).

The area of excursion between limb segments during SLS is indicative of joint laxity, strength, coordination, balance or a combination of the aforementioned that may affect knee joint stability during standing (16,18). The ROLS Symmetry Index (SI) (%) is the ratio between lower limb ROLS values expressed as a percentage where 100% suggests absolute symmetry (16,18). The lower the ROLS SI value, the greater the asymmetry, with the lower limb having the larger ROLS value demonstrating greater excursion. Although a small sample size was noted, differences in ROLS SI were found when comparing ROLS values calculated from a single sensor placed at the sacrum to ROLS values calculated from both knee sleeves and lower limbs in a group of student-athletes undergoing rehabilitation for an MSK injury (16). The ROLS SI was used to identify lower limb stability and postural deficits between collegiate women basketball players who were and were not injured during the respective season (18). Lastly, significant difference in

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ROLS SI was found post-concussion in a group of student athletes compared to standard clinical measures of balance (19).

Given the high incidence of injury and re-injury in the athletic population there is a need to develop a predictive measure for risk of injury that is clinically friendly, quick to administer, and accurate. The purpose of the study is to determine if a measure of lower limb segment stability derived from body-worn inertial measurement unit (IMU) can predict relative risk for lower limb musculoskeletal injury among Division I Collegiate Football Players (D1CFP).

Methods

The study was approved by the University of Miami Institutional Review Board (IRB). The study examined three different cohorts over the course of three years. Each cohort was followed for 1 year.

Study Procedures

Informed consent, anthropometric and demographic data (height, weight, position, age, and year in school) were collected from 104 subjects over the course of three consecutive seasons. Subjects were tested within the month prior to the start of their football season. Subjects wore standard indoor physical training gear (T-shirt, shorts, socks, and sneakers) with the instrumented knee sleeves and performed SLS indoors on a hardwood gymnasium floor (16).

Prior to administration of the SLS test, sensors were calibrated and aligned using an iPad Air 2 tablet with a customized mobile application (16). After alignment was complete, subjects

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performed a 30-second SLS trial on each lower limb with a 60-second rest between trials. Trials and testing session data were stored automatically within the device and uploaded to a secure cloud-based server. Those subjects who were healthy during the pre-season but acquired an injury that required surgery during the current season was classified as the "injured group". Those subjects who were healthy during the pre-season and did not acquire an injury during the current season were classified as the "non-injured group". The subjects were not re-classified after completion of their season.

Musculoskeletal injuries requiring surgery to the subjects were tracked and recorded by the team athletic trainers during the season. To maintain confidentiality, subjects were de-identified within the database.

Outcome Measures

ROLS and ROLS SI

The ROLS is a measure of lower limb stability developed to quantify bi-directional excursion of the thigh and shank segments with IMU sensors during a 30-second SLS test (16). Maximal excursion of the thigh and shank, in the anterior or posterior direction and medial or lateral direction, used to calculate the ROLS value (cm^2) (Figure 1(b)) The ROLS Excursion Diagram (ED) is a graphic representation of the segmental excursion. Lower limb ROLS values ≥ 10 cm^2 have been found to be indicative of the presence of an underlying lower limb injury and greater instability following concussion, respectively (18,19). The ROLS Symmetry Index (SI)

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(%) is the ratio between lower limb ROLS values expressed as a percentage where 100% suggests absolute symmetry (16,18,19). The ROLS SI is calculated as follows:

$$\text{ROLS SI (\%)} = 100 - \left|100 \times \left(\frac{ROLS_L - ROLS_R}{ROLS_L + ROLS_R}\right)\right|$$

where $ROLS_L$ (cm^2) is the ROLS value measured during left SLS and $ROLS_R$ (cm^2) is the ROLS value measured during right SLS. The ROLS SI values ≤ 80% have been found to be indicative of the presence of an underlying lower limb injury and greater instability following concussion, respectively (18,19).

Statistical Analysis

Data analysis was performed on IBM SPSS Statistical Software version 22. The ROLS values for the left and right lower limbs and ROLS SI were calculated for all subjects at preseason. Those subjects that experienced a lower limb injury that resulted in surgery were labeled as "injured" and those that did not have an injury were labeled as "non-injured." A student t-test was performed to examine the differences in ROLS values for each lower limb and ROLS SI between the injured and non-injured subjects. Receiver operating characteristic (ROC) curve analysis was performed to determine the diagnostic accuracy of ROLS and ROLS SI in discriminating between those subjects who were injured or non-injured (20). Area under the curve (AUC) was calculated for ROLS and ROLS SI to determine their ability to identify subjects at risk for MSK injury requiring surgery. The AUC values were evaluated based on the following criteria: 1.0 = perfect test; 0.9–0.99 = excellent test; 0.8–0.89 = good test; 0.7–0.79 = fair test; and 0.51–0.69 = poor test (20). ROC values were examined for differences between limbs and subjects were flagged if

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ROLS SI values were less than 80%. Frequency analysis and confusion matrix was developed to describe the preliminary ROLS injury classification based on the following criteria: ROLS value > 10 cm^2 for one or both lower limbs and ROLS SI < 80%. Within the confusion matrix, true positive, true negative, false positive false negative, accuracy ((true positive + true negative)/total), error rate ((false positive + false negative)/total), sensitivity (true positive/actual yes), specificity (true no/actual no), and error rate ((false positive + false negative)/total). False positive was classified as those non-injured athletes that had ROLS value >10 cm^2 for either one or both limbs and/or <80% ROLS SI (18). False negative was classified as those injured subjects who had ROLS value <10 cm^2 for the injured limb and/or a ROLS SI value >80% (18). Statistical significance was < 0.05.

Results

All 104 subjects were healthy (i.e., able to participate in full workouts without modifications) at the time of testing and were able to perform SLS on both lower limbs for 30 seconds. The mean ± standard deviation (range) height, weight, and age for the study sample was 186.69 ± 7.52 cm (167.64 – 203.20), 103.47 ± 19.68 kg (65.77 – 154.22), and 19.47 ± 1.14 years old (19-23), respectively. Of the 104, 70 were classified as non-injured and 34 were classified as "injured". Table 1 provide a description of the subjects by position. Running back (RB), wide receivers (WR), defensive back (DB), and linebacker (LB) produced the highest number of surgeries. The majority of injured subjects were either freshman, red-shirt freshman, sophomore, or red-shirt sophomores (55%). Table 2 provides a description and frequency of the lower limb

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injuries sustained during the study period, with the most prevalent injuries being to the anterior cruciate ligament, meniscus, and medial collateral ligament. Within the injured group, eight subjects had injuries to multiple lower limb structures.

Table 3 describes the baseline ROLS and ROLS SI values for both groups. The right and left lower limb ROLS values for the non-injured group were 6.94 ± 6.0 cm^2 and 7.10 ± 5.50 cm^2, respectively. No difference in ROLS values were found between the right and left lower limb for the non-injured group ($p = 0.69$). Because there were no difference between the right and left lower limb ROLS values for the non-injured group, we decided to use the right lower limb ROLS values to compare to the injured limb of the injured group and the left lower limb values to the non-injured limb of the injured group. Significant differences in ROLS values was found between the injured and non-injured lower limbs for the injured group (11.40 ± 7.12 cm^2 and 9.36 ± 17.36 cm^2, $p = 0.001$). The ROLS value of the injured lower limb in the injured group was significantly higher than the right lower limb in the non-injured group ($p = 0.003$). The mean ± SD ROLS SI was 82.86 ± 14.75% and 66.41 ± 16.35% for the non-injured and injured group, respectively, with significant differences noted ($p < 0.001$).

The ROC curve for the ROLS value comparing the injured and non-injured limb is illustrated in Figure 2. The AUC was 0.72 ($p < 0.001$; 95% CI: 0.62 - 0.83) with a standard error (SE) of 0.05 indicating that ROLS value alone has fair diagnostic accuracy in detecting those D1CFPs at risk for lower limb injury requiring surgery.

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The ROLS SI, which combines the ROLS values for both lower limbs, demonstrated an AUC of 0.8 ($p < 0.001$; 95% CI: 0.71 - 0.88) with a SE of 0.04 (Figure 3). The results indicate that the ROLS SI has good diagnostic accuracy in detecting those healthy D1CFPs at risk for lower limb injury resulting in surgery.

A confusion matrix was developed to describe the preliminary injury classification using ROLS values and ROLS SI (Table 4). The classification criteria for injury risk was the following: ROLS value > 10 cm^2 for one or both lower limbs and ROLS SI < 80%. Thirty-one subjects were classified as True Positive because the met the above criteria and were injured. Sixty-four subjects were classified as True Negative because they did not meet the above criteria and were not injured. Of the 70 subjects who were non-injured, six false positive (0.09) would have been incorrectly identified as being at risk for injury resulting in surgery based on the criteria of having a ROLS greater than 10 cm^2 of either limb and a ROLS SI < 80. Similarly, of the 33 subjects who had surgery, only two would have been identified as having a false negative result. The ROLS injury classification demonstrating an accuracy of 0.92. The sensitivity of the ROLS injury classification was 0.94. The specificity of the ROLS injury classification was 0.91. The error rate of the ROLS injury classification was 0.08.

Discussion

We believe this is the first study incorporating a mobile sensor system in conjunction with a performance-based outcome measure to determine risk for lower limb MSK injury in healthy D1CFPs. Our results suggest that subjects who had an in-season injury requiring surgery had significantly higher ROLS values on the affected limb and greater asymmetry between limbs in preseason screening.

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The study results demonstrated that the ROLS values had fair accuracy, in identifying those subjects at an increased risk of injury that required surgery. Perhaps of greater significance were the study results indicating that the D1CFP with lower ROLS SI scores are at greater risk of injury than those with high ROLS SI values. The AUC value of 0.8 suggests that the use of ROLS SI which accounts for both lower limbs to determine injury risk in our subjects had greater accuracy than using just the ROLS value alone (18). Of note, by combining the ROLS values for both lower limbs and the ROLS SI, during preseason testing, the percentage of those classified as false positive and false negative was 9% (n = 6) and 6% (n = 2). These findings are consistent with numerous studies which have demonstrated the relationship between limb asymmetry and injury risk (12,14,23,24).

The present study involved several limitations that should be considered in future efforts aimed at injury prediction. First, we only tested D1CFPs, whether or not similar findings occur in other populations (different sports, gender, etc.) is unknown at this time. Second, the subjects presented with different lower limb MSK injuries therefore targeting multi-structural vs. isolated injuries was not possible (e.g., ACL and meniscal injury vs. isolated ACL injury). Third, student athletes were only tested once a year, right before the beginning of their fall practice and season. Fourth, the mechanism of injury (e.g., contact vs non-contact, specific activity being performed, etc.) and the environment (e.g., , practice vs. competition, surface, etc.) in which the injury occurred was not documented at the time of injury. Fifth, prior MSK injuries to the lower limbs were not documented at baseline testing. Future studies should account for prior significant (e.g., requiring rehabilitation with or without surgery) MSK injuries to the lower limbs in a larger sample of subjects.

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It may be of greater benefit to test the student's athletes immediately before and after the season and periodically during their training and conditioning season because neuromusculoskeletal changes occur with age, intervention, and exposure to different training and sport activities. The practice of multiple planned data collection per year could allow the rehabilitation team to identify emergence of an impairment or limitation prior to the occurrence of a lower limb injury. In addition, investigators should consider identifying postural stability in those who were injured but did not require surgery, as the authors believe there may be a protective stability that may mediate the severity of an injury. Lastly, the ability to use repeated measures throughout the rehabilitation following lower limb injury to longitudinally track change over time and potentially help guide the rehabilitation intervention may prove useful in establishing post-injury return to play criteria.

Conclusion

This study demonstrated the efficacy of an IMU-based mobile system for the direct assessment of lower limb stability and risk for injury resulting in surgery in DICFPSs. Increase in thigh and shank excursions and/or decrease in symmetry index (SI) may predict risk of injury to the less stable limb. Higher predictive ability was achieved comparing lower limbs rather than single leg assessments. The ROLS is a measure that demonstrated good sensitivity and specificity in detecting those collegiate D1CFPs at risk for lower limb MSK injuries requiring surgery that were healthy and asymptomatic during preseason testing. The technologies used in this study were shown to be both feasible to implement and applicable to the population being tested. By combining the metrics used herein with additional outcome measures, a clinician may be able to gather a holistic idea of the potential causes for postural instability.

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Figure Captions

Figure 1. (a) Thigh and shank movements of the supporting limb during SLS (blue solid lines). The red dashed line is a perimeter of each segmental excursion. The green dotted line and black dash-dot line are maximum excursions of the thigh and shank in the ML and AP direction, respectively. (b) A border is the ROLS excursion diagram (ED). The excursion values for the AP and ML excursion for the thigh and shank are combined to determine to excursion of the lower limb, and presented as the ROLS ED. The area of the ROLS ED is the ROLS value (in cm^2).

Figure 2. The ROC Curve for ROLS value between the injured group (n=34) and the non-injured group (n = 70).

Figure 3. The ROC Curve for ROLS SI value between the injured group (n = 34) and the non-injured group (n = 70).

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Figure 1

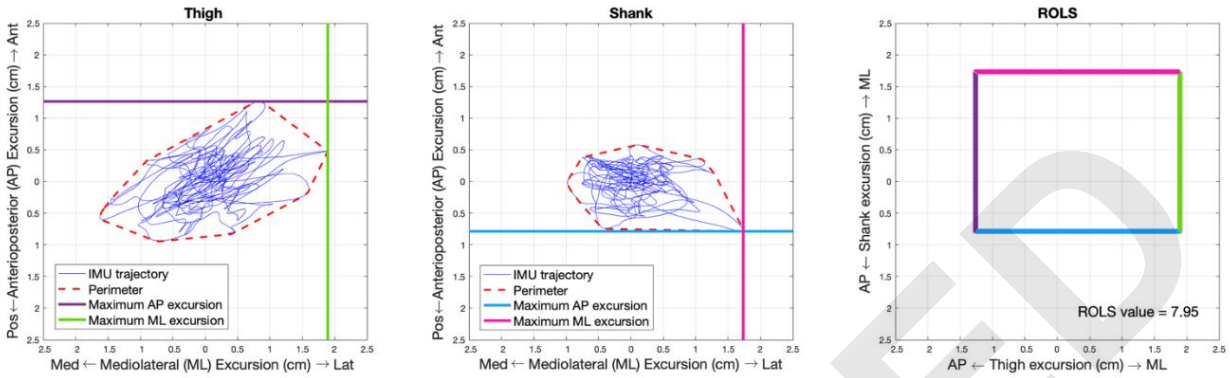

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Figure 2

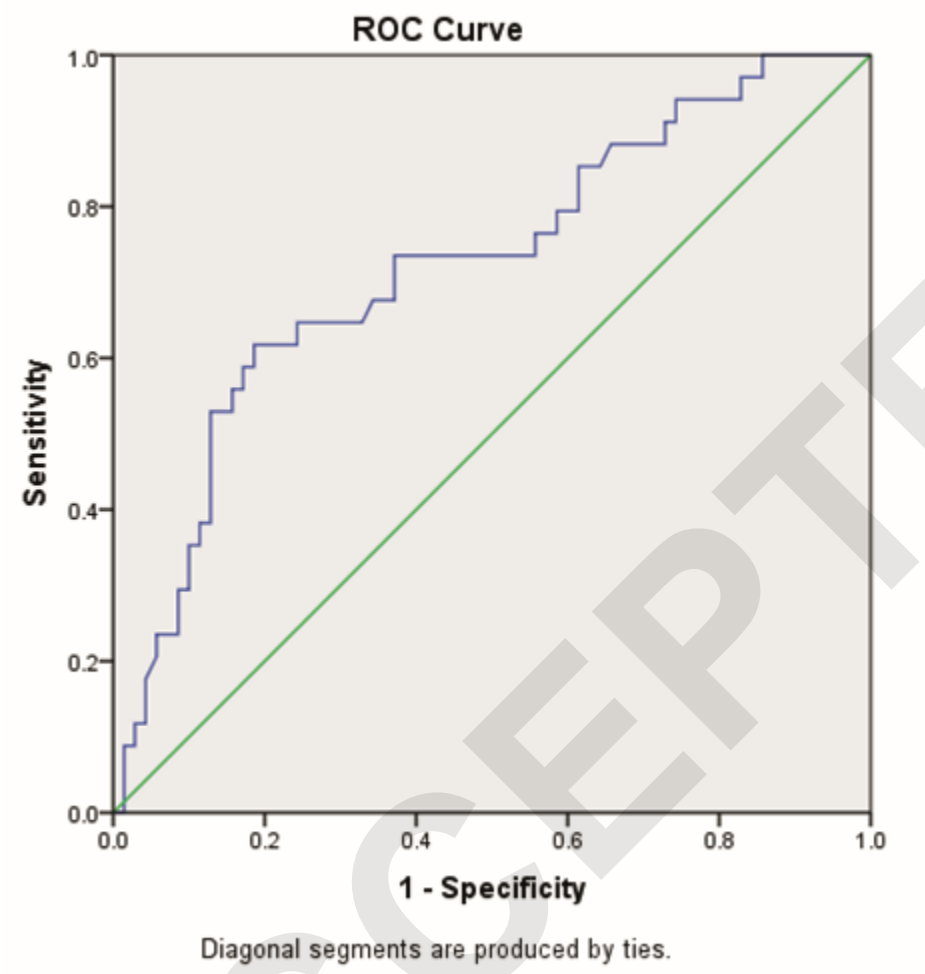

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Figure 3

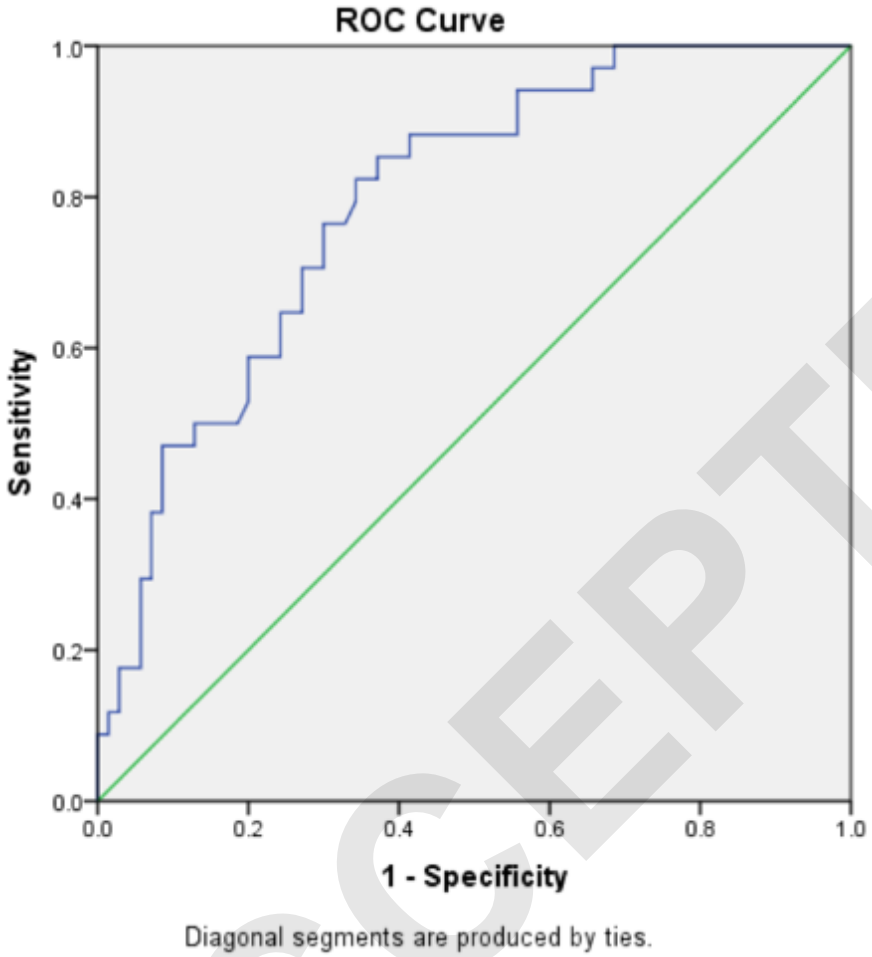

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Table 1. Summary of subjects based on football position and injury.

	DL n (%)	LB n (%)	DB n (%)	OL n (%)	QB n (%)	RB n (%)	FB n (%)	TE n (%)	WR n (%)	P n (%)	K n (%)	Total n (%)
Non-injured	8(73)	9 (69)	14 (64)	12 (69)	4 (80)	3 (37)	3 (75)	5 (100)	8 (53)	1 (100)	3 (100)	70(68)
Injured	3(27)	4 (31)	7 (33)	5 (29)	1 (20)	6 (63)	1(25)	0 (0)	7 (47)	0 (0)	0 (0)	34(32)
Total	11(11)	13(13)	21(20)	17(17)	5(5)	9(8)	4(4)	5(5)	15(15)	1(<1)	3 (3)	104

DL = Defensive Line; LB = Linebacker; DB = Defensive Back; OL = Offensive Line; QB = Quarterback; RB = Running Back; FB = Fullback; TE = Tight End; WR = Wide Receiver; P = Punter; K = Kicker

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Table 2. Description and frequency of lower limb body structures that suffered injuries for the surgical subjects.

Body Structures	***Frequency***
Anterior Cruciate Ligament	14
Meniscus	7
Medial Collateral Ligament	4
Ankle Ligament Tear	3
Ankle Fracture	3
Articular Knee Cartilage	2
Patella Osteochondritis Dissecans	2
Lis Franc Fracture	2
Metatarsal Fracture	2
Os Trigonum Syndrome	1
Avulsion Fracture (Semitendinosus)	1
Knee Dislocation/Nerve Injury	1
Lumbar Discectomy (Unilateral)	1
Medial Patellofemoral Ligament Tear	1
Popliteofibular Ligament Tear	1
Femoral Condyle Fracture	1
Hip flexor Tendon tear	1
Quadriceps Tendon Tear	1
Hamstring Tear	1
Knee Posterior Capsule Tear	1

Within the surgical group, eight subjects had injuries to multiple lower limb structures.

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Table 3. Descriptive values for ROLS and ROLS between limbs for those non-injured and injured subjects.

	Non-injured Group mean ± SD n = 70	**Injured Group mean ± SD n = 34**	**Non-injured vs Injured** p-value
Right Limb/Injured Limb ROLS (cm^2)	6.94 ± 6.0	11.40 ± 7.12	**0.003**
Left Limb/Non-Injured Limb ROLS (cm^2)	7.10 ± 5.50	9.36 ± 17.36	0.46
ROLS SI (%)	82.86 ± 14.75	65.58 ± 16.46	**< 0.001**
Right Limb/Injured vs Left Limb/Non-injured p-value	0.69	**0.001**	

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Table 4. Injury Classification Confusion Matrix Model for Region of Limb Stability

n = 104	**Predicted: NO**	**Predicted: YES**	
Actual: NO	TN = 64	FP = 6	70
Actual: YES	FN = 2	TP = 31	33
	66	37	

TP = True Positive; TN = True Negative; FP = False Positive/Type I error; FN = False Negative/Type II error

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